



Computer System and Architecture

Machine Problem

MIDTERM

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List the steps that you perform to come up with the program.

Step 1: Planning the sections.

Every NASM program needs .data (initialized text that are needed), .bss (empty buffers that are needed), and .text (code) so i started by deciding what goes where.

Step 2: Write strings in .data.

I used db (define byte) for each string and ended every line with 10, the ASCII newline, so the terminal moves down. I gave each string a label like title and a length using title_len equ \$ - title. \$ is the current address, so subtracting the label gives the byte count.

Step 3: Make ASCII art and borders.

I stored banners, borders, and prompts as separate strings. Indent prompt labels (p1, p2, etc.) with spaces so the colons line up when printed. I kept the naming conventions as short and easy to remember.

Step 4: Reserve buffers in .bss.

I used resb 100 for each input field (b1 through b18), reserving 100 bytes of empty memory to hold the user's answer which is a bit of a limit/bottleneck for my program.

Step 5: Making the two helper functions.

printString uses syscall 4 (sys_write): put 4 in EAX, 1 (stdout) in EBX, the string address in ECX, the length in EDX, then int 0x80, and ret at the end. PrintString basically prints the initialized stings.

readString uses syscall 3 (sys_read): put 3 in EAX, 0 (stdin) in EBX, the buffer in ECX, 100 in EDX, then int 0x80. Add ret. It reads the input and then i instructed it to store.

Step 6: Build _start top to bottom.

Set global _start, then for each line, i used a pretty simple pattern/convention: mov ecx, label, mov edx, label_len, call printString. For each prompt: print the prompt, then mov ecx, buffer, mov edx, 100, call readString.

Step 7: Group into sections.

I just grouped them by adding spaces and comments. Banner, title, then three prompts. Print "My First" art and three prompts. Repeat for "My Faves" and "Hobbies." Add dividers between sections.

Step 8: Exit cleanly.

End with mov eax, 1, xor ebx, ebx, int 0x80 to return status zero.



Source Code:

MIDTERM EXAM

[illegible]



```
art18_len equ $ - art18

art19 db "  |_|  |_|\\_, | |_|  \\_,_| \\/ \\_||_|_", 10
art19_len equ $ - art19

art20 db "          _/ |                                ", 10
art20_len equ $ - art20

art21 db "          |_|/                                ", 10
art21_len equ $ - art21

;Hobbies
art22 db "  _  _  _  _  _                                ", 10
art22_len equ $ - art22

art23 db "  | | | |  | | | |  ( )                                ", 10
art23_len equ $ - art23

art24 db "  | |_| | _ | |_| | |_| _ _ _ _ ", 10
art24_len equ $ - art24

art25 db "  | _  |/_ \\| '_ \\| '_ \\| |/_ \\_ \\_", 10
art25_len equ $ - art25

art26 db "  | | | | ( ) | | ) | | ) | |  _/\\_ \\", 10
art26_len equ $ - art26

art27 db "  |_|  |_|\\_/_/|_._/|_._/|_|\\_||_|_", 10
art27_len equ $ - art27

;border design
```



```
border      db '+-----'
+', 10
border_len  equ $ - border

thin_line   db '|-----'
|', 10
thin_len    equ $ - thin_line

;spacer
blank       db '|'
|', 10
blank_len   equ $ - blank

;prompts
p1 db '    Name: '
p1_len equ $ - p1

p2 db '    Email: '
p2_len equ $ - p2

p3 db '    Blog/Website: '
p3_len equ $ - p3

p4 db '    First Achievement: '
p4_len equ $ - p4

p5 db '    First Risk: '
p5_len equ $ - p5

p6 db '    First Happy Time: '
p6_len equ $ - p6
```



```
p7 db '    Color/s: '
p7_len equ $ - p7

p8 db '    Perfume: '
p8_len equ $ - p8

p9 db '    Music: '
p9_len equ $ - p9

p10 db '    Singer: '
p10_len equ $ - p10

p11 db '    Song: '
p11_len equ $ - p11

p12 db '    Food: '
p12_len equ $ - p12

p13 db '    Weekend Activity: '
p13_len equ $ - p13

p14 db '    TV Shows: '
p14_len equ $ - p14

p15 db '    Movie: '
p15_len equ $ - p15

p16 db '    Book: '
p16_len equ $ - p16

p17 db '    Celebs: '
p17_len equ $ - p17
```



```
p18 db '    Role Model: '
p18_len equ $ - p18

newline db 10
newline_len equ $ - newline

section .bss
    b1 resb 100
    b2 resb 100 ;;99 characters, the last if for hte newline
    b3 resb 100
    b4 resb 100
    b5 resb 100
    b6 resb 100
    b7 resb 100
    b8 resb 100
    b9 resb 100
    b10 resb 100
    b11 resb 100
    b12 resb 100
    b13 resb 100
    b14 resb 100
    b15 resb 100
    b16 resb 100
    b17 resb 100
    b18 resb 100

section .text
    global _start

_start:
    ;About Me
```



```
mov ecx, art1
mov edx, art1_len
call printString

mov ecx, art2
mov edx, art2_len
call printString

mov ecx, art3
mov edx, art3_len
call printString

mov ecx, art4
mov edx, art4_len
call printString

mov ecx, art5
mov edx, art5_len
call printString

;top border
mov ecx, border
mov edx, border_len
call printString

;blank line inside box
mov ecx, blank
mov edx, blank_len
call printString

;title
mov ecx, title
```



```
mov edx, title_len
call printString

;blank line inside box
mov ecx, blank
mov edx, blank_len
call printString

;divider
mov ecx, thin_line
mov edx, thin_len
call printString

;input section
mov ecx, p1
mov edx, p1_len
call printString
mov ecx, b1
mov edx, 100
call readString

mov ecx, p2
mov edx, p2_len
call printString
mov ecx, b2
mov edx, 100
call readString

mov ecx, p3
mov edx, p3_len
call printString
mov ecx, b3
```



```
mov edx, 100
call readString

;divider
mov ecx, thin_line
mov edx, thin_len
call printString

;border before My First art
mov ecx, border
mov edx, border_len
call printString

;My First ASCII art
mov ecx, art6
mov edx, art6_len
call printString

mov ecx, art7
mov edx, art7_len
call printString

mov ecx, art8
mov edx, art8_len
call printString

mov ecx, art9
mov edx, art9_len
call printString

mov ecx, art10
mov edx, art10_len
```



```
call printString

mov ecx, art11
mov edx, art11_len
call printString

mov ecx, art12
mov edx, art12_len
call printString

mov ecx, art13
mov edx, art13_len
call printString

;border after My First art
mov ecx, border
mov edx, border_len
call printString

mov ecx, p4
mov edx, p4_len
call printString
mov ecx, b4
mov edx, 100
call readString

mov ecx, p5
mov edx, p5_len
call printString
mov ecx, b5
mov edx, 100
call readString
```



```
mov ecx, p6
mov edx, p6_len
call printString
mov ecx, b6
mov edx, 100
call readString

;divider
mov ecx, thin_line
mov edx, thin_len
call printString

;border before My Faves art
mov ecx, border
mov edx, border_len
call printString

;My Faves ASCII art
mov ecx, art14
mov edx, art14_len
call printString

mov ecx, art15
mov edx, art15_len
call printString

mov ecx, art16
mov edx, art16_len
call printString

mov ecx, art17
```



```
mov edx, art17_len
call printString

mov ecx, art18
mov edx, art18_len
call printString

mov ecx, art19
mov edx, art19_len
call printString

mov ecx, art20
mov edx, art20_len
call printString

mov ecx, art21
mov edx, art21_len
call printString

;border after My Faves art
mov ecx, border
mov edx, border_len
call printString

mov ecx, p7
mov edx, p7_len
call printString
mov ecx, b7
mov edx, 100
call readString

mov ecx, p8
```



```
mov edx, p8_len
call printString
mov ecx, b8
mov edx, 100
call readString

mov ecx, p9
mov edx, p9_len
call printString
mov ecx, b9
mov edx, 100
call readString

mov ecx, p10
mov edx, p10_len
call printString
mov ecx, b10
mov edx, 100
call readString

mov ecx, p11
mov edx, p11_len
call printString
mov ecx, b11
mov edx, 100
call readString

mov ecx, p12
mov edx, p12_len
call printString
mov ecx, b12
```



```
mov edx, 100
call readString

mov ecx, p13
mov edx, p13_len
call printString
mov ecx, b13
mov edx, 100
call readString

;divider
mov ecx, thin_line
mov edx, thin_len
call printString

;border before Hobbies art
mov ecx, border
mov edx, border_len
call printString

;Hobbies ASCII art
mov ecx, art22
mov edx, art22_len
call printString

mov ecx, art23
mov edx, art23_len
call printString

mov ecx, art24
mov edx, art24_len
call printString
```



```
mov ecx, art25
mov edx, art25_len
call printString

mov ecx, art26
mov edx, art26_len
call printString

mov ecx, art27
mov edx, art27_len
call printString

;border after Hobbies art
mov ecx, border
mov edx, border_len
call printString

mov ecx, p14
mov edx, p14_len
call printString
mov ecx, b14
mov edx, 100
call readString

mov ecx, p15
mov edx, p15_len
call printString
mov ecx, b15
mov edx, 100
call readString
```



```
mov ecx, p16
mov edx, p16_len
call printString
mov ecx, b16
mov edx, 100
call readString

mov ecx, p17
mov edx, p17_len
call printString
mov ecx, b17
mov edx, 100
call readString

mov ecx, p18
mov edx, p18_len
call printString
mov ecx, b18
mov edx, 100
call readString

;exit
mov eax, 1
xor ebx, ebx
int 0x80

printString:
mov eax, 4 ;; sys_write
mov ebx, 1 ;; stdout
int 0x80
ret
```



```
readString:
    mov eax, 3 ;; sys_read
    mov ebx, 0 ;; stdin
    int 0x80
    ret
```

Output:

```

  A B O U T M E
+-----+
|           |
|   by Vladimir Gonzales   |
|           |
+-----+
| Name: Vladimir Gonzales  |
| Email: vagonzales@fit.edu.ph |
| Blog/Website: vagonzales.com.ph |
|           |
+-----+
  M Y F O R E
+-----+
| First Achievement: graduating school |
| First Risk: betting on sports       |
| First Happy Time: vacation with family |
|           |
+-----+
  M Y F O R E
+-----+
| Color/s: black and white |
| Perfume: anything dior  |
| Music: r&b, pop, hiphop  |
| Singer: iv of spades     |
| Song: come inside of my heart |
| Food: pizza              |
| Weekend Activity: sleeping |
|           |
+-----+
  H I S T O R Y
+-----+
| TV Shows: none           |
| Movie: avengers          |
| Book: none               |
| Celebs: benedict cumberbatch |
| Role Model: my parents    |
+-----+
```



Link to video explanation:

<https://drive.google.com/drive/folders/1Pb4d69Qmy3VwDc87KBtYxwLi0QLelzqm?usp=sharing>